

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
UNIVERSITY QUESTION BANK

CST463 WEB PROGRAMMING (2019 Scheme)

(CS368 WEB TECHNOLOGIES – 2015 Scheme)

MODULE I – PART 1 (WWW)

KTU Syllabus and Model Question Paper (2019 Scheme)

1. Define WWW. List any two examples of web server & web browser. Differentiate between URL and a domain? (3)
2. What is MIME? Give the expansion of MIME. List four examples for MIME types. State the reason why MIME type specification is necessary in a request-response transaction between a browser and server.

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3. List out the roles of web servers in web technology. (3)
4. Write the syntax of URL? Explain with one Example? (3)
5. What do you mean by MIME? (4)
6. Explain the request and response phases in HTTP? (5)

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7. Why do you call MIME as an extension feature? Justify with suitable statements. (3)
8. What is the role of query string in web communication? Write an example for a query string. (3)
9. What is the difference between web servers and browsers in the context of their implementation and functionality? (4)
10. Describe the situations in which POST method is preferred and why? (4)

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11. What is a MIME type? Give examples. Why should MIME type information be essentially included in HTTP responses? (3)

12. Mention any three HTTP methods. Provide a brief description about the functionality of each. (3)
13. Write down the general format of an HTTP request and an HTTP response. What is the purpose of the following HTTP headers? Also, identify whether they are included with an HTTP header/response or both. (4)
 - i. host
 - ii. last-modified

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14. What are MIME specifications? What is its utility in the internet? Write the MIME specifications for (a) a jpeg image (b) a plain text document (3)
15. State the format of an URL. Explain the concepts of domain names and host names with reference to URLs. (3)
16. List any five well known web servers. Briefly describe the operation of a web server. (5)
17. State and describe the general structure of an HTTP header and an HTTP response. What is the function of the following HTTP methods. (5)
 - (i) GET
 - (ii) PUT
 - (iii) POST

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18. Write the syntax of URL with suitable example. (3)
19. What is the significance of HTTP? (3)
20. What is MIME type? Give examples. Why should MIME type information be essentially included in HTTP responses? (3)

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21. Explain the role of web servers in web technology. (3)
22. Explain URL with an example. (3)
23. What is MIME? Describe how is it useful. (4)
24. Write HTTP request and response message structure. (4)

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25. What is the significance of domain names in Internet. (3)
26. Write and explain the general format of URL. (3)
27. Explain the sequence of steps performed during HTTP communication. (5)

28. What is the purpose of using MIME format specifier? Explain with one eg.. (4)

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29. What is a URL? Explain the different parts of URL with example. (3)

30. Explain HEAD, PUT and POST HTTP methods. (3)

31. Describe the use of MIME by explaining the various type and subtypes. (4)

32. Explain the HTTP web communication process. (5)

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MODULE I – PART 2 (HTML)

KTU Syllabus and Model Question Paper (2019 Scheme)

1. Design a webpage that displays the following table. (6)

Food Item	Recommended Intake			
	age<15		age>15	
	gm	Kcal	gm	Kcal
Cerials	1000	2000	750	1760
NonCerials	450	800	350	600

2. Construct a valid HTML document for your personal Profile registration page for a Job Site www.123Jobs.com. Add relevant HTML elements in a table, to accept a minimum of 10 different fields which includes your name, address, phone, email address, your picture, your college; your branch, fields for your personal history (Minimum 3 fields), favourite theory and practical subjects (Checkbox), Username, Password (password).
3. Create a static HTML document for your portfolio, which includes the following contents: your name, address, Mobile Number and email address. Also add the details about your college, university, your major and the batch of study. Include a picture of yourself and at least one other image (friend/pet/role model) to the document with a short description about that. Add three paragraphs about your personal history, with links to your social media profile. Also create an ordered list for describing your Skill Set & an unordered list showing your Strengths & Weaknesses. (8)
4. What is the difference between radio buttons and checkboxes when implemented using HTML? Write HTML code to implement a form which has the following elements:
 - i. A textbox which can accept a maximum of 25 characters
 - ii. Three radio buttons with valid Label, Names and values
 - iii. Three check boxes buttons with valid Label, Names and values
 - iv. A selection list containing four items, two which are always visible
 - v. A submit button clicking on which will prompt the browser to send the form data to the server "http://www..mysite.com/reg.php" using "POST" method and reset button to clear its contents. You can use any text of your choice to label the form elements.

(8)

5. Write the equivalent HTML code to implement the following in a web page: (6)
 - (i) An image titled “birds.jpg” with a height of 100 pixels and width of 200 pixels. If the image cannot be accessed, a message “No image available” should be displayed
 - (ii) A hyperlink to the URL “www.mysite.com/birds.jpg”. The hyperlink should have the label “Click Here”.
6. What is codec? Recognize the role of controls attribute in <video> & <audio> tag in HTML. Use the COVID vaccination promotional video ‘MySafety.mp4’ in a web page with suitable HTML code, ‘autoplay’ option enabled and displayed in a standard dimension 750X500.

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7. Explain the different types of buttons in HTML (3)
8. Write notes on headings and block quotations. (4)
9. Write HTML code to design a web page to create a table with 6 rows and 3 columns for entering the mark of 5 students. Assume suitable headings for each column. (4)

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10. How can you differentiate header row and data rows in an HTML table? Illustrate with an example. (3)
11. Describe the structure of an HTML document with an example. Give description for the major tags used. (5)
12. How can you give an alternate text for an image so that text is displayed when the image cannot be loaded? Show with an example. (4)
13. Design a table in HTML to display the marks of 5 students for 3 subjects. Student name should be the row headings and subject name should be the column headings. Marks can be of your choice. (5)

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14. Differentiate between block tags and inline tags. Which tag preserves the white spaces in its content? Illustrate with suitable examples. (3)
15. Write the equivalent HTML code to implement the following in a web page:

- (i) An image titled “birds.jpg” with a height of 100 pixels and width of 200 pixels. If the image cannot be accessed, a message “No image available” should be displayed
 - (ii) A hyperlink to the URL “www.mysite.com/birds.jpg”. The hyperlink should have the label “Click Here”.
16. Write down any four HTML tags and their purpose with suitable examples. What are tag attributes? (4)
17. What is the difference between radio buttons and checkboxes when implemented using HTML? Write HTML code to implement a form which has the following elements:
- i. A textbox which can accept a maximum of 25 characters.
 - ii. Three radio buttons.
 - iii. A selection list containing four items, two which are always visible.
 - iv. A submit button, clicking on which will prompt the browser to send the form data to the server “http://responses.mysite.com” using “POST” method.
- You can use any text of your choice to label the form elements. (5)
18. Differentiate between “rowspan” and “colspan” table attributes. Write HTML code to create a table whose structure is as below:

19. Insert any text of your choice in the table cells. The first row is considered to be containing a heading. The background colour for the table should be red, and the text in the cells should all be centre aligned. (5)

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20. Differentiate between block tags and inline tags with suitable examples for each. (3)
21. Write HTML code to display a table with the following structure. You can fill in the table cells with content of your choice. (4)

22. Write equivalent HTML code to realize the following. (4)
- Display a paragraph of text in bold, red color and in italics
 - Display an image termed “birds.jpg” with height 100 pixels and width 200 pixels and in the middle of the page; if the image cannot be accessed, a message “Image Not Found” should be displayed
23. Write HTML code to create a web form to capture customer feedback in an online shopping site. The form should have the following elements. (5)
- Text boxes to enter customer first name and last name
 - radio buttons to enter gender
 - drop down list with entries (1,2,..5) to provide a rating
 - a multiline text area to provide detailed feedback (limited to 200 characters) and
 - a button titled “Submit”.
- When the submit button is pressed the data entered in the form should be sent to the url – “feedback.mysite.com” using POST method. (5)

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24. Write an HTML code to display the following table. (3)

Flower Show	
Flower Name	Colour
Red Rose	Red
White Rose	White
Jasmine	

25. Write an HTML code to display your class time table? (6)
26. Write the equivalent HTML code to implement the following in a web page: (3)
- 1) An image titled “flower.jpg” with proper attribute to set height, width and message text.
 - 2) Unordered list with values tea, coffee and milk
27. Explain following html tags with proper example. (Any Four) (4)
1. <input>
 2.
 3. <dl>
 4. <form>
 5. <a>
28. Differentiate between block tags and inline tags with suitable examples. (2)

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29. Explain the importance of alt attribute for tag. (3)
30. Explain image hyperlink with an example. (3)
31. Explain <blockquote> tag with an example. (2)
32. Design a webpage that display the table: (5)

Food Item	Recommended Intake			
	age<15		age>15	
	gm	Kcal	gm	Kcal
Cerials	1000	2000	750	1760
NonCerials	450	800	350	600

33. Design an HTML page that display the list: (5)

- Animals
 - Vertebrates
 1. Warm Blooded
 2. Cold Blooded
 - Invertebrates
 1. With Jointed Legs
 2. Without Legs
- Plants
 - Flowering Plants
 - Non-Flowering Plants

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34. Explain the tags <pre> and <blockquote> in HTML with examples. (3)
35. Write HTML code for the following with some sample data. (3)
36. Display an image which is a link to another document. (3)
37. Display an unordered list of 4 items. (3)
38. Display a subscript to a variable in HTML. (3)
39. Write down the syntactic difference between HTML and XHTML. (5)
40. Design an HTML document that has a form with the following controls: (9)
(i) A text box to collect Students' name.
(ii) Four checkboxes, one each for the following items:
i. Diploma ii. B.Tech iii. M.Tech iv. Phd
(iii) Drop down menu for selecting Academic year.
(iv) A collection of three radio buttons that are labeled as follows:
i. Visa ii. Master Card iii. Debit Card

- (v) Submit and Reset buttons.
Give suitable labels for each item.

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33. Distinguish between cell spacing and cell padding attributes of table tag with a suitable example. (3)
34. Write the code for the following output: (5)

Price List

	Veg	Non Veg	Snacks
Breakfast	30	85	15
Lunch	45	90	20
Dinner	40	65	15

35. Demonstrate the use of size and multiple attributes for select tag with help of code. Draw the relevant output also. (4)
36. Explain basic structure of HTML document. (5)
37. Write HTML code to implement a form which has the following elements: (4)
- A textbox which has label "NAME" accept a maximum of 30 characters.
 - A textbox which has label "PASSWORD" accept a maximum 8 characters and it mask characters when user enters
 - A selection list containing four items, "CS201", "CS202", "CS203", "CS204", first two of which are always visible
 - A submit button, clicking on which will prompt the browser to send the form data to the server "http://responses.mysite.com" using GET method.

MODULE II – PART 1 (CSS)

KTU Syllabus and Model Question Paper (2019 Scheme)

1. Discuss the various CSS style sheet levels with suitable examples. How are conflicts resolved when multiple style rules apply to a single web page element? (3)
2. List the order of precedence of style levels. Organize a sample web page for providing 'KTU BTech Honours Regulation 19' for KTU and use embedded Style sheet to apply minimum 5 styles for list, tables and pages. (6)
3. Organize a sample web page for the event 'Raagam2021' at your campus and use embedded Style sheets to apply a minimum 5 styles. State the Style Specification format of embedded style sheets.
4. Write CSS style rules to implement the following in a web page:
 - a. to display the content of hyperlinks with yellow background color and in italics
 - b. to display the contents of unordered lists in bold and in Arial font
 - c. to display a background image titled "birds.jpg" with no tiling.
5. What are different ways of adjusting spacing in a text with suitable example. (3)
6. Write CSS and the corresponding HTML code for the following: (6)
 - i. Set the background color for the hover and active link states to "green"
 - ii. Set the list style for unordered lists to "square".
 - iii. Set "Flower.png" as the background image of the page and set 3% margin for the pages
 - iv. Set dashed border for left and right and double border for top & bottom of a table with 2 rows.

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7. What do you mean by "class" and "id" in CSS? Explain with example? (3)
8. How to link an external style sheet to an HTML document? What are the attributes required for that? (3)
9. What is bootstrap? What is the advantage of using bootstrap? (4)
10. Write CSS code for the following: (5)
 - i) set the background color for the hover and active link states to "yellow".
 - ii) Set the list style for unordered lists to "square".

- iii) Set "Flower.gif" as the background image of the page
 - iv) Set dashed border for the document.
11. Differentiate between and <div> tags with examples. (4)

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12. What is the specialty of pseudo selectors compared to other selector forms? (3)
13. What are the different ways of adjusting spacing in a text? Explain the various style attributes used for that. (3)
14. Write CSS code for setting the following styles. (6)
- i) Table - A header row with background colour and bold font and other rows without that.
 - ii) Unordered list – Set a smiley image as bullet.
15. What are the advantages of using a CSS framework? Explain the features of Bootstrap in relation to the above answer. (4)
16. External style sheets are usually preferred over other kinds of style sheets. Why? (4)

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17. Differentiate between the action of span and div tags with suitable examples. (3)
18. Write CSS style rules to implement the following in a web page: (3)
- a. to display the content of hyperlinks with yellow background colour and in italics
 - b. to display the contents of unordered lists in bold and in Arial font
 - c. to display a background image titled "birds.jpg" with no tiling.
19. Discuss the various CSS style sheet levels with suitable examples. How are conflicts resolved when multiple style rules apply to a single web page element? (4)
20. Distinguish between descendant selectors and child selectors in CSS. If there is a style rule given by `body p { color : red; font-family : courier; }` and another style rule `body > p { color : green; }`, what styles will apply to the content of a paragraph in the web page body? (4)
21. What are class selectors in CSS? Suppose that it is required to display the text content of p tags in two different styles depending on the context.

Style 1 – text should be displayed in blue colour, right aligned, underlined, with font style of italics and spacing between the words of text set to 2 cm.

Style 2 – text should be displayed with a background colour of yellow, having a dashed, blue colour border, and with a padding of 1.5 cm

Write the equivalent CSS style rules. (5)

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22. Illustrate the use of and <div> tags with suitable examples.(3)
23. Distinguish between class selectors and generic selectors for specifying CSS rule with suitable examples. (3)
24. Write CSS rules to enforce the following styles in a web page: (4)
 - i. Paragraph text should always be right aligned and in bold, and should be surrounded by a black, solid border 5 pixels thick
 - ii. Table header text should always be underlined, in italics style and also in Times New Roman font; table cell contents should have a yellow background, be centre aligned and have a padding of 10 pixels
25. Illustrate the difference between the effect of descendant selectors and child selectors when specifying CSS style rules with examples. If an HTML document has the style rules `body {color:blue; font-family:Arial}` and `p{color:red}` specified, what will be the style applicable to the contents of a p element? Why? (5)
26. Briefly describe the different levels of CSS style sheets. Explain the precedence rules applicable for style rules defined at various CSS levels with an example. (4)

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27. Write about CSS Box model with neat diagram? (3)
28. Explain how to control spacing in between text, words and lines with proper examples (3)
29. How to link an external style sheet to an HTML document? Describe the various attributes used? (3)
30. Write CSS code for the following: (4)
 - i) set the background color for the hover and active link states to "yellow".
 - ii) Set the list style for ordered lists to "lower case alphabet".
 - iii) Set "Boat.gif" as the background image of the page.

- iv) Set dotted border for the document.
- 31. What is a selector in CSS? What do you mean by contextual selector and where it is used? (3)
- 32. Explain following tags with proper example? 1. 2.<div> (2)

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- 33. Write a CSS rule that makes all text 1.5 times larger than the base font of the system and colors the text red. (3)
- 34. Add an embedded style sheet to an HTML document that should contain a rule that displays h1 elements in blue. In addition, create a rule that displays all links in blue without underlining them. (3)
- 35. Explain different levels of style sheets with examples. (6)
- 36. Explain box model of elements in a web document. (5)
- 37. Explain class selector and id selector with examples. (4)

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- 38. Differentiate generic selectors and universal selector with examples. (3)
- 39. What do you mean by Box Model ?(3)
- 40. Create an external style sheet which will make styles as follows.
 - i) create a class named, big, which set any four font properties.
 - ii) When mouse pointer moves to the content of <h2>, it should be displayed as red.
 - iii) Unordered list with bullet type disc.
- 41. Design an HTML document also which will make use of this style sheet. (5)
- 42. Differentiate <div> and tags (2)
- 43. What are the advantages of Bootstrap? (2)
- 44. Write styles for elements as follows (5)
 - i) table with border type dotted and color blue.
 - ii) <p> element with background color red and text underlined.
 - iii) set both line spacing and padding to 2 px in a <p> element.
 - iv) set background image to 'plane.png' with size 200X300 pixels.
 - v) display an image 'sky.png' on the right side of the display and the text flow around the left side of the image.

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45. Differentiate between id selector and generic selector in CSS with suitable examples. (3)
46. What are the uses of text-indent and text-align properties? What are the possible values the properties can take? Demonstrate with example. (3)
47. How are conflicts resolved when there are two different values for the same property on the same element in a document.(4)
48. Explain different levels of style sheets with example. (4)
49. What is Box Model? By choosing a suitable example explain margin, border and padding properties. (4)

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MODULE II – PART 2 (JAVASCRIPT)

KTU Syllabus and Model Question Paper (2019 Scheme)

1. Write the code for an HTML document with embedded JavaScript scripts, which initially displays a paragraph with text "Welcome" and a button titled "Click". When the button is clicked, the message "Hello from JavaScript" in bold should replace the paragraph text
2. Illustrate the implementation of a JavaScript function greeting () using external .js file, to display a welcome message, when you click on a Button in an HTML page. (3)
3. Illustrate the different ways of Array declaration in JavaScript. Describe the function of the following JavaScript Array object methods with examples.
(i) join (ii) slice (8)
4. Illustrate the usage of JavaScript DOM in event handling and explain any three methods with example. (8)

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5. What are the different methods by which we can convert a string to a number in JavaScript? (3)
6. Explain various String properties and methods present in JavaScript. (5)
7. Write a JavaScript program to find the factorial of a number. (4)
8. Write a JavaScript code to solve a quadratic equation by reading the coefficients through dialog box. Also use confirm dialog box to check whether user wants to continue or not. (5)

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9. What is a callback function in JavaScript? State how it is different from normal functions. (3)
10. What is the difference between JavaScript primitives and objects in the context of how they are stored in memory? (3)
11. Write a JavaScript program to implement various mathematical operations. Input should be prompted for and confirmation should be asked before proceeding to next operation. (5)

12. Illustrate with suitable code samples, the different ways of Array declaration in JavaScript. Arrays in JavaScript are said to be more flexible, why? (5)

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13. List three methods for creating an array object in JavaScript. What function does the array methods join() and slice() perform? (3)
14. What is meant by DOM? How is it useful for JavaScript scripts? (3)
15. How are event handlers registered in JavaScript? Write the code for an HTML document with embedded JavaScript scripts, which initially displays a paragraph with text "Welcome" and a button titled "Click". When the button is clicked, the message "Hello from JavaScript" in bold should replace the paragraph text. (5)
16. What are the output statements provided by JavaScript? Write embedded JavaScript code which displays a dialog box that prompts the user to enter a string. The script should then concatenate the string after the string "Hello " and then print the result 5 times in italics one below the other in the web page. (4)

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17. Differentiate between client-side and server-side. Mention any one benefit of client-side scripting over server-side scripting. (3)
18. Write JavaScript code to accept a number from the user and then displays the factorial of the same. Use a function to calculate and return the factorial value. (3)
19. Describe the concepts of events and event handlers in JavaScript. Suppose that it is required to create a blank web page having only a single button titled "Click Here". When the button is clicked, a centre-aligned message "Welcome to JavaScript" should be displayed in bold, green colour. Write HTML and JavaScript code to implement the same. (5)
20. Explain the concept of wrapper objects in Javascript. What is the difference between the Javascript statements `var num = new Number("12")` and `var num = Number("12")`? (4)
21. Describe the function of the following JavaScript Array object methods with examples – (i) join (ii) shift (iii) slice (3)

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22. Write a JavaScript program to read an integer value using prompt and display its square? (3)
23. What is the syntax of JavaScript and Why JavaScript is enclosed in HTML comments? (3)
24. Discuss various types of control statement in JavaScript? (4)
25. JavaScript is known as object-based language. Justify your answer? (2)
26. What is client-side scripting? Write a JavaScript program which accepts N as input and print first N even numbers. (5)
27. Explain how to create objects in JavaScript with example? (4)

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28. Explain sort() method in JavaScript with examples. (3)
29. Explain array creation in JavaScript with examples. (3)
30. Explain various String properties and methods present in JavaScript. (4)
31. Write JavaScript program to find factorial of a number. Use prompt dialog box to get the input from user and use confirm dialog box to check whether user wants to continue or not. (5)

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32. Write a JavaScript that prompt the user to enter a number and print the sum of the digits of that number in the document. (3)
33. Explain the implicit and explicit type conversions in JavaScript. (3)
34. Explain the various methods associated with array objects in JavaScript. (4)
35. Write a JavaScript that will compute the mean and median of an array of numbers. Separate functions should be written for each computation. (5)

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36. With suitable code snippets, explain alert(), confirm and prompt() methods.(3)
37. With suitable figure, distinguish between primitives and objects.(3)
38. Explain DOM concept in JavaScript.(5)
39. Write an HTML code and JavaScript program to find factorial of number. The JavaScript should be an external file. (5)
40. Write a JavaScript program to demonstrate various String functions such as

- i. `length()`
- ii. `charAt()`
- iii. `substring()`
- iv. `toLowerCase()`.

The input string should be given using `prompt()` function. (5)

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MODULE III (PHP Basics)

KTU Syllabus and Model Question Paper (2019 Scheme)

1. During the process of fetching a web page from a web server to a client browser, at what point does an embedded PHP script get executed. What are the two modes that the PHP processor operates in? Explain (6)
2. Why is PHP considered to be dynamically typed? Distinguish between implode and explode function in PHP with suitable examples. (8)
3. Write a PHP program to compute the sum of the positive integers up to 100 using do while.
4. Write an embedded PHP script which displays the factorial of all numbers from 1 to 10 in a table in the web page. The factorial should be calculated and returned from a function. The table headings should be "Number" and "Factorial"
5. Build a PHP script to check whether a given number is prime or not. (3)
6. Explain any six string handling functions used in PHP with example. (6)
7. How does a PHP array differ from an array in C? List the different ways to create an array in PHP with an example. Explain any 4 functions that deals with PHP array. (8)
8. Write a PHP program to store the name and roll no of 10 students in an Associative Array and Use foreach loop to process the array and perform asort, rsort and ksort in the array. Illustrate with suitable output data
9. Write equivalent PHP statements corresponding to the following: (8)
 - (i) Declare an associative array named "ages" to store the key-value pairs ("Alice", 30), ("Bob", 30), ("Harry", 35), ("Mary", 32).
 - (ii) Modify the value associated with the key "Mary" to 28.
 - (iii) Sort the array according to values maintaining the key-value relationships and print the sorted key-value pairs.
 - (iv) Delete the entry identified by the key "Bob".

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10. What are the different ways to create an array in PHP? Explain with example? (4)
11. What is server-side scripting? Write a PHP program to check whether the given number is Armstrong or not? (6)
12. Write a PHP script to count the instances of words in a string? (5)

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13. What are the primitives supported by PHP? (4)
14. Write a PHP script to search for a particular string pattern in a text. (5)
15. Illustrate with a sample PHP script, how functions are implemented in PHP. (5)

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16. During the process of fetching a web page from a web server to a client browser, at what point does an embedded PHP script get executed? Can a client see the PHP script in the retrieved page? How does this differ from JavaScript script execution? (3)
17. What are the two modes that the PHP processor operates in? Explain. (2)
18. How is the type of a variable determined in PHP? Distinguish between bound and unbound variables. How can a variable be tested to check whether it is bound? (3)
19. What is the purpose of the explode() and implode() functions in PHP? Illustrate with suitable examples. (2)
20. Write an embedded PHP script which displays the factorial of all numbers from 1 to 10 in a table in the web page. The factorial should be calculated and returned from a function. The table headings should be "Number" and "Factorial". (5)
21. What are associative arrays? How are associative arrays declared in PHP? Illustrate with an example. (2)
22. Briefly describe, with suitable examples, the various methods for iterating over the contents of a PHP array. (3)

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23. Why is PHP considered to be dynamically typed? How is it possible to check whether a variable in PHP has been assigned a value? (2)
24. Distinguish between implode and explode function in PHP with suitable examples. (3)
25. Write a PHP function which checks whether a string passed to it starts with the character pattern "PHP". (5)
26. Write equivalent PHP statements corresponding to the following: (4)
 - i. Declare an associative array named "ages" to store the key-value pairs ("Alice", 30), ("Bob", 30), ("Harry", 35), ("Mary", 32).

- ii. Modify the value associated with the key “Mary” to 28.
- iii. Sort the array according to values maintaining the key-value relationships and print the sorted key-value pairs.
- iv. Delete the entry identified by the key “Bob”.

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- 27. How to do pattern Matching in PHP? (4)
- 28. Write a PHP script to count the instances of words in string? (6)
- 29. What are the different ways to create array in PHP? Explain with example? (4)
- 30. List out any 4 string functions in PHP? (2)

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- 31. Explain how to use functions in a PHP program with an example. (3)
- 32. Explain any five string functions used in PHP. (5)
- 33. Explain different methods to create Array in PHP with examples. (6)
- 34. Write a PHP program to check whether a number is prime number or not. (5)
- 35. How does a PHP array differ from an array in C? Explain any 4 functions that deals with PHP array. (5)
- 36. Write a PHP script to count the instances of words in a string. (2)

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- 37. Explain the logical internal structure of an array in PHP. (5)
- 38. Explain the local, static and global variables in PHP with one example. (5)
- 39. Write a PHP function that will find out the frequency of occurrence of each word in a constant string given. Print the words with odd frequencies in the order of occurrence.

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- 40. Describe the various ways in which Arrays are created in PHP. (4)
- 41. Explain the concept of Scope and Lifetime of a PHP variable. (5)
- 42. PHP uses dynamic typing. Justify. (5)
- 43. What are the methods for sequential access of Array elements in PHP? (5)
- 44. Explain sort, asort, ksort and its reverse functions in PHP. (5)

MODULE IV (PHP Forms, Cookies, DB Connectivity)

KTU Syllabus and Model Question Paper (2019 Scheme)

1. Describe how input from an HTML form is retrieved in a PHP program, with an example. (3)
2. Design an HTML page which enters a given number, write a PHP program to display a message indicating, whether the number is odd or even, when clicking on the submit button.
3. Design the HTML page which enters a given number and embed the PHP code to display a message indicating, whether the number is odd or even, when clicking on the 'CHECK NUMBER' button. (6)
4. Create a valid HTML document for yourself, including your name, address, and email address. Also add your college; your major and the course. Perform form handling in PHP and process the output using POST method.
5. Discuss the various steps for establishing PHP-MySQL connection with a MySQL database? (3)
6. Write a PHP form handling program to verify the user authentication credentials of a web page using MySQL connection and store the userid value as a Session variable if the userid is valid.
7. Write a PHP form handling program to perform the user registration of any website with a minimum of 5 different fields and insert the data into a MySQL table after establishing necessary connections with the DB. (8)
8. What are the uses of cookies in web pages? Describe syntax for setting cookies in PHP. How can you access and delete the cookie using setcookie() function? (6)

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9. Design an HTML form for entering a number by the user. Write a PHP code to display a message indicating, whether the number is odd or even, when clicking on the submit button. (5)
10. What is the significance of cookies in web? How can a cookie be created and destroyed in PHP? (5)

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11. Design a simple HTML form to input a string and to display whether it is palindrome or not when the form is submitted. Use a PHP script. (6)
12. What are the contexts in which session tracking can be helpful? (5)
13. How does the use of cookies help in server-side scripting? Give examples for data stored using cookies. (5)
14. What is the function of cookies? Explain why session tracking is sometimes a better alternative to using cookies. (5)

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15. What is the purpose of the implicit arrays \$_POST and \$_GET in PHP? Consider that a web page displays a form containing two text boxes (named num1 and num2), where the user enters numeric data. Write a PHP script which collects this form data, finds the sum, difference and the product of the two numbers and then displays the same with suitable messages. Assume that the script is to be embedded in the web page specified by the action attribute of the form and that the method used when the form is submitted is GET. (5)

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16. What are benefits of using session tracking over cookies? (3)
17. Write HTML code to create a form that prompts users to enter a number (n) as well as his/her name (name), and also has a Submit button. Also, write a PHP script which will be invoked in the server when the form is submitted. The script should cause the message "Hello <name>" to be displayed n number of times with one message per line. (6)
18. Discuss the benefits and drawbacks of cookies. Write notes on the PHP mechanisms for setting cookies, as well as for retrieving values stored in cookies. (4)

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19. Design a simple HTML form to input a string and to display whether it is Armstrong or not when the form is submitted. Use a PHP script. (5)
20. What is the significance of cookies in web? How can a cookie be created and destroyed in PHP?

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21. Describe how input from an XHTML form is retrieved in a PHP program, with an example. (5)
22. What are the uses of cookies in web pages? Describe syntax for setting cookies in PHP. (3)

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23. Write the HTML code to provide a form that collects the name and marks in five subjects of a student. Write a PHP script that will process these details and print the total and average marks of the student. (6)
24. Explain the session tracking in PHP with one example. (4)
25. What is the purpose of using cookies? How can a cookie be created in a PHP script? How can a script determine whether a particular cookie exists?

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26. What is a cookie? How does PHP support the concept of Cookies? (5)

MODULE V - PART 1 (JSON)

KTU Syllabus and Model Question Paper (2019 Scheme)

1. List the datatypes used in JSON with examples.
2. Explain the use of parse () and stringify() functions in JSON with examples.
3. Describe the schema of a document implemented in JSON with suitable examples. (3)
4. Enumerate the data types in JSON. Illustrate the document definition of a 'Student document' using JSON Schema. (8)

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5. Describe the schema of a document implemented in JSON. (6)
6. List the various data types used in JSON. (5)

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7. What are the various data types used in JSON? Explain the use of parse() and stringify() functions in JSON with examples. (4)

MODULE V - PART 2 (LARAVEL)

KTU Syllabus and Model Question Paper (2019 Scheme)

1. What is Route Model Binding in Laravel? Which types of route model binding are supported in Laravel?
2. Explain how laravel performs route handling using routes calling controller methods?
3. Explain the role of Resource controllers in Laravel. (3)
4. With a neat diagram, explain about Laravel MVC Framework. (6)
5. Discuss in detail about Laravel's Routing mechanisms. (8)
6. Discuss the following in Laravel Views (6)
 - i. Creating & Rendering Views
 - ii. Passing Data to Views
 - iii. Sharing Data with All Views
